

An AI-Driven Decision Support System for Tomato Cultivation in Sri Lanka

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Project Proposal Report

FARTHAS A L M – IT22190734

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Department of Software Engineering
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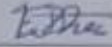
An AI-Driven Decision Support System for Tomato Cultivation in Sri Lanka

(AI-Based Tomato Leaf Disease Detection Using CNN and Explainable AI)

DECLARATION

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
Farthas ALM	IT22190734	

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature
Brind Gunapala	12/03/2026	

ABSTRACT

Tomato (*Solanum lycopersicum*) is a commercially significant horticultural crop in Sri Lanka, contributing approximately LKR 127 million in export value in 2022, with production reaching approximately 88,000 metric tons [1]. However, tomato cultivation is increasingly threatened by a range of leaf diseases, including Early Blight (*Alternaria solani*), Bacterial Spot (*Xanthomonas campestris*), and Tomato Leaf Curl Virus — all of which are recognized as major constraints to yield and crop quality. A critical challenge in building AI-driven diagnostic systems for these diseases is that publicly available tomato leaf disease datasets, such as PlantVillage, contain only single-disease-labeled images. In real agricultural conditions, however, a tomato leaf may simultaneously exhibit multiple diseases. This dataset limitation makes direct multi-label classification approaches infeasible.

To address this research gap, this study proposes a novel parallel binary classification architecture comprising three independent Convolutional Neural Network (CNN) models — one dedicated to each target disease. Each model independently produces a binary detection output (disease present / not present) along with a confidence score. When all three models process a leaf image simultaneously, the combined outputs enable multi-disease detection without requiring multi-label training data. Explainability is incorporated through Gradient-weighted Class Activation Mapping (Grad-CAM), which generates visual heatmaps that highlight the leaf regions most influential in each model's prediction, thereby improving transparency and farmer trust. An advisory module further enhances system utility by retrieving verified treatment guidelines from the Department of Agriculture, Sri Lanka, and employing a Large Language Model (LLM) exclusively to translate these recommendations into clear, accessible farmer-friendly language. The system is deployed through a React Native mobile application backed by a Node.js server, making disease diagnosis accessible to smallholder farmers across all agro-climatic zones of Sri Lanka.

Keywords: Tomato disease detection, Convolutional Neural Network, binary classification, Explainable AI, Grad-CAM, Early Blight, Bacterial Spot, Tomato Leaf Curl Virus, mobile application, advisory system

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
CNN	Convolutional Neural Network
XAI	Explainable Artificial Intelligence
Grad-CAM	Gradient-weighted Class Activation Mapping
LLM	Large Language Model
DoA	Department of Agriculture
HORDI	Horticultural Crops Research and Development Institute
ML	Machine Learning
DL	Deep Learning
API	Application Programming Interface
JSON	JavaScript Object Notation
SDK	Software Development Kit
SDLC	Software Development Life Cycle
UAT	User Acceptance Testing
EB	Early Blight
BS	Bacterial Spot
TLCV	Tomato Leaf Curl Virus
SL	Sri Lanka
MT	Metric Tons

1. INTRODUCTION

Sri Lanka's economic and social fabric is deeply interwoven with agriculture, and within this sector, horticulture stands out as a critical driver of rural income and food self-sufficiency. Within the horticulture sub-sector, the tomato (*Solanum lycopersicum*) occupies a position of considerable strategic value. According to the Department of Agriculture Sri Lanka, tomatoes can be cultivated across almost all agro-climatic zones of the country, with the exception of the up-country wet zone, and are recognised as an important cash crop with export potential [2]. The crop is valued for its nutritional richness in vitamins A and C and forms a daily dietary staple.

Sri Lanka's tomato sector is an important commercial horticulture segment. IndexBox market data valued tomato production at approximately USD 127 million in export price in 2022, with a harvested area of around 5,900 hectares and an average yield of 15 tons per hectare [1], [21].

According to FAOSTAT data, tomato production in Sri Lanka reached 74.2 kt in 2024 — a 7.68% decline from the previous year — having peaked at an all-time high of 101 kt in 2018 [20]. Despite this recent contraction, Sri Lanka ranks 87th among 168 countries in tomato production globally, underscoring the crop's significance to national horticulture. The sector faces persistent threats from plant diseases, which significantly reduce yield, quality, and farmer income.

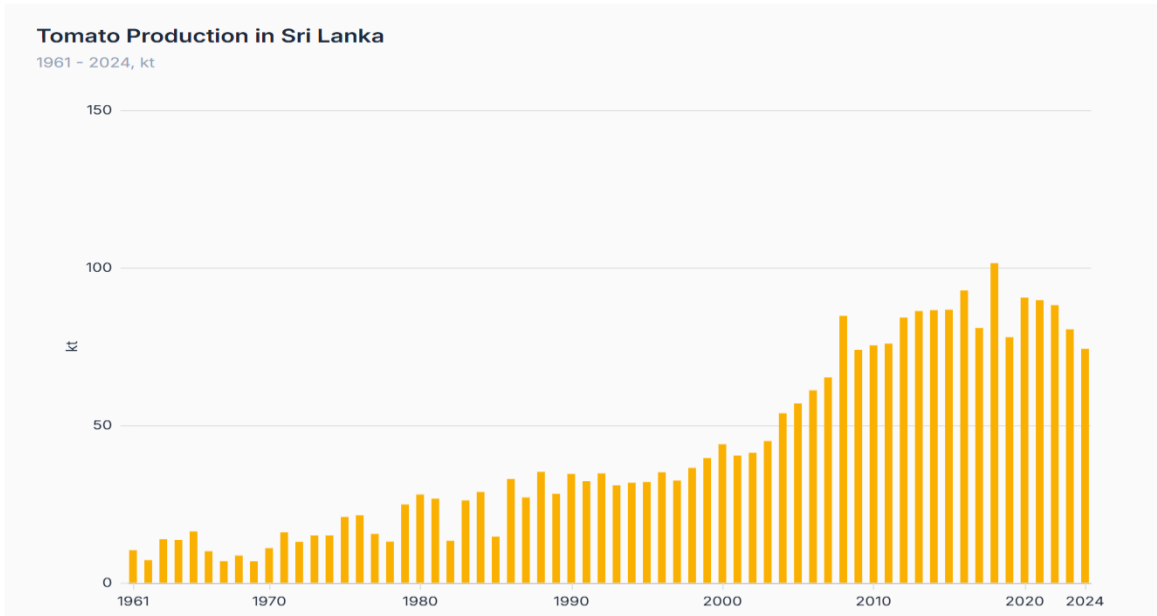


Figure 1.1: Tomato Production in Sri Lanka, 1969–2024 (Source: FAOSTAT via Helgi Library [20])

Source: [Tomato Production in Sri Lanka | Helgi Library](#)

Traditional diagnostic approaches depend on physical examination and specialist consultation — methods that are both resource-intensive and susceptible to error, particularly among farming communities with limited access to expert support [3], [22].

Within the domain of computer vision, Convolutional Neural Networks (CNNs) have emerged as a powerful instrument for automated plant pathology diagnosis, offering image-based detection capabilities that were previously unattainable [12].

However, a critical structural limitation exists in publicly available tomato leaf disease datasets: images are labeled with only one disease per leaf, even though in practice a single tomato leaf may be simultaneously infected by multiple diseases. This single-label constraint prevents direct multi-label training and necessitates an architectural innovation to enable multi-disease detection from single-disease training data. This research proposes a solution through a parallel binary classification architecture — three independent CNN models, each specialised in detecting one target disease — that when executed simultaneously on a single leaf image, produces a combined multi-disease detection output without requiring multi-label ground truth.

A key barrier to widespread AI adoption in agricultural settings is the opacity of model decision-making. Without the ability to interrogate or validate predictions, end users such as farmers and field officers are unlikely to place sufficient confidence in automated outputs to act upon them. To address this, Explainable AI (XAI) techniques — specifically Gradient-weighted Class Activation Mapping (Grad-CAM) — are integrated into the system to generate visual heatmaps that highlight the specific leaf regions that influenced each model's prediction. These visual explanations make the AI's reasoning transparent and actionable [7], [8].

Complementing detection and explainability, the proposed system incorporates an advisory module that retrieves verified treatment and management guidelines from the Department of Agriculture Sri Lanka's HORDI resources [2] and uses a Large Language Model (LLM) solely to convert these verified recommendations into clear, farmer-friendly language. The entire system is delivered through a React Native mobile application, ensuring accessibility for smallholder farmers who rely on smartphones for agricultural information [17].

1.1 Background and Context

Tomato cultivation in Sri Lanka spans both Maha and Yala agricultural seasons across multiple agro-climatic zones. The Department of Agriculture's Horticultural Crops Research and Development Institute (HORDI) has developed and released over thirteen tomato varieties suited to Sri Lankan growing conditions, including Thilina, Ravi, Maheshi, and HORDI Tomato Hybrid 03, among others indicating the diverse geographic and climatic coverage of tomato production across the island.

Despite production growth, tomato farmers face consistent challenges from plant diseases. The three diseases targeted in this study — Early Blight, Bacterial Spot, and Tomato Leaf Curl Virus — are among the most prevalent and economically damaging. Early Blight presents with characteristic dark lesions arranged in concentric rings, predominantly affecting older foliage and advancing towards younger growth. Bacterial Spot is marked by small, initially water-saturated spots that enlarge into irregular darkened patches across the leaf surface. Tomato Leaf Curl Virus induces characteristic upward rolling of leaves, chlorotic discolouration, and suppressed plant growth, with whiteflies serving as the principal transmission vector. The visual similarity of early-stage symptoms across these diseases makes accurate field diagnosis particularly challenging without technological support.

This diagnostic challenge is compounded in Sri Lanka by the insufficiency of agricultural advisory infrastructure in rural areas, where extension officer coverage relative to the farming population falls well short of what is required. In the absence of accessible expert support, most farmers resort to experiential judgement or make time-consuming trips to district offices, during which period disease progression continues unchecked. AI-powered smartphone tools offer a compelling solution to this gap, enabling near-instantaneous on-site disease identification that significantly shortens the interval between symptom onset and corrective action. Furthermore, the challenge of detecting multiple diseases simultaneously in a single leaf image has been demonstrated in related crop domains; for instance, Asha et al. showed that DenseNet121 with sigmoid activation and optimised threshold achieved 98.6% accuracy for multi-disease identification in citrus leaf images [19], validating the architectural approach of using independent binary classifiers for multi-disease detection.

From an AI architecture perspective, the core innovation of this research addresses a well-documented dataset limitation. Publicly available plant disease datasets, including PlantVillage — which is the most widely used benchmark in plant disease detection research — contain images labeled with a single disease per image [3]. This characteristic stems from the controlled inoculation

procedures used during data collection, which do not reflect the multi-pathogen co-occurrence that routinely arises under natural growing conditions. Attempting to train a direct multi-label classifier on this type of data is methodologically untenable, given the complete absence of co-labelled training instances. The proposed parallel binary classifier architecture is designed specifically to overcome this limitation.

This research directly contributes to SDG 2 (Zero Hunger) by improving food security through reduced crop losses, SDG 9 (Industry, Innovation and Infrastructure) through the application of AI for agricultural advancement, and SDG 15 (Life on Land) by promoting sustainable land use through precision disease management.

1.2 Literature Review

1.2.1 Deep Learning for Plant Disease Detection

The groundwork for image-based plant disease classification using deep learning was laid by Mohanty, Hughes, and Salathé (2016), who demonstrated that a CNN trained on approximately 54,000 PlantVillage samples spanning 14 crops and 26 conditions could attain 99.35% accuracy, positioning PlantVillage as the primary reference dataset for the community [3].

Kibriya and Rafique established the suitability of CNN architectures for tomato pathology classification, showing that hierarchical feature extraction enables recognition of subtle textural and structural cues associated with specific disease conditions [4]. Their findings further emphasised that training data quality and augmentation are critical prerequisites for robust generalisation.

Despite these contributions, notable limitations persist across these studies. Mohanty et al. [3] reported that the same model achieved only 31.4% accuracy on uncontrolled field images, exposing a critical gap between laboratory performance and real-world deployment reliability. Furthermore, neither Kibriya and Rafique [4] nor Garba and Jain [5] extended their work beyond classification to include explainability features or mobile deployment, and none of these systems were designed for or validated within the Sri Lankan agro-climatic context.

1.2.2 Transfer Learning and Benchmark Architectures

In agricultural AI applications, transfer learning has established itself as an indispensable strategy, allowing models pre-conditioned on large repositories such as ImageNet to be repurposed for specialised tasks with substantially smaller domain-specific datasets [14].

In an evaluation of competing CNN configurations for leaf disease identification, Padol and Yadav found that both architectural choice and training protocol exert considerable influence on predictive

accuracy and computational efficiency — factors of particular relevance when targeting mobile deployment environments [9].

Despite the promise of transfer learning, existing implementations in tomato disease detection share a critical limitation: none have been extended into a deployable, farmer-facing mobile application. Mahmud et al. [7] achieved notable accuracy figures — 0.993 using EfficientNetB3, 0.94 with Xception, and 0.93 with MobileNetV2 — yet did not integrate these models into a real-time mobile platform, nor was the system contextualised for Sri Lankan crop varieties, field conditions, or agricultural advisory services. This gap between research prototype and practical deployment remains unaddressed in the existing literature.

1.2.3 Explainable AI in Agricultural Disease Detection

The growing application of AI to consequential domains including precision agriculture and clinical care has brought model interpretability to the forefront of research discourse [18].

As formalised by Selvaraju et al., Grad-CAM derives class-discriminative localisation maps by computing the gradient of the target class score with respect to activations in the final convolutional layer, producing a spatial importance map that identifies prediction-relevant image regions [7], [11].

While Grad-CAM has been validated as an effective interpretability tool in research settings, existing studies have not integrated it into a mobile-accessible interface for agricultural users. The work of Drake and Henry [8] explicitly notes that most XAI deployments in agriculture have not been validated through user studies with actual farmers, leaving a significant gap between theoretical explainability and practical farmer trust. No prior study delivers real-time Grad-CAM visualisations directly to smallholder farmers through a smartphone application in the Sri Lankan context.

1.2.4 Multi-Disease Detection from Single-Label Datasets

Widely adopted benchmark repositories, including PlantVillage — the most referenced collection in the field — assign a single disease category to each image, reflecting the controlled inoculation procedures used during data collection rather than the multi-pathogen co-occurrence common in natural field environments [3].

Conventional approaches to multi-label classification depend on the availability of training samples in which individual instances carry simultaneous annotations across multiple classes — a condition that single-label datasets, by design, cannot fulfil. A viable architectural workaround involves deploying an ensemble of independent binary detectors, each trained to recognise a single disease class. When these models are executed concurrently on the same input, the system can infer multi-

disease presence from their combined outputs — circumventing the need for multi-label annotations and avoiding the class distribution and ambiguity challenges that arise when a single model is tasked with producing multi-class outputs from single-label data.

Empirical evidence in adjacent plant pathology domains has confirmed that this binary ensemble strategy is capable of achieving robust multi-disease identification even when trained exclusively on single-label data. Asha et al. [19] investigated multi-disease identification in single citrus leaf images, where leaves were simultaneously infected with up to three diseases — Anthracnose, Melanose, and Bacterial Brown Spot. Rather than relying on multi-label annotations, they applied a sigmoid activation layer with an optimised classification threshold, achieving 98.6% accuracy using DenseNet121. This outcome offers compelling support for the architectural premise of the current research, confirming that co-infection detection without multi-label supervision is achievable and motivating the adoption of an analogous parallel binary CNN framework for tomato disease identification.

Nevertheless, the work of Asha et al. [19] has two key limitations in relation to the present study. First, it was conducted on citrus crops and has not been replicated for tomato, which presents a distinct set of pathological markers and disease co-occurrence patterns. Second, the study did not incorporate a mobile deployment component or an advisory output layer, meaning the detection capability was not translated into actionable guidance for end users. These gaps directly inform the design of the proposed system.

1.2.5 Mobile and Advisory Systems for Agricultural AI

Within the Sri Lankan agricultural context, Jayasinghe et al. designed an intelligent platform for crop disease identification and geographic spread forecasting, affirming that AI solutions can be effectively tailored to the operational realities of smallholder farmers [10]. A key insight from that work was that classifying a disease in isolation is insufficient; practical value is greatly amplified when diagnostic outputs are paired with actionable management guidance.

An emerging paradigm in agricultural AI involves coupling predictive models with language-based advisory components. Rather than tasking a language model with generating agronomic advice from its own parameters — an approach prone to hallucination and factual distortion — a more reliable design employs the LLM strictly as a communication bridge that reformulates pre-validated content from authoritative sources, such as DoA guidelines, into language accessible to farmers of varying literacy levels.

The work of Jayasinghe et al. [10] represents the closest existing system to the proposed research within the Sri Lankan context. However, it targets rice and vegetable crops rather than tomato, does

not incorporate multi-disease detection, lacks Grad-CAM visual explainability, and does not link its outputs to DoA-verified management advisories. These absences collectively constitute the motivation for the current research and define its contribution relative to the state of the art.

1.3 Research Gap

The limitations identified across the reviewed literature collectively delineate a set of critical research gaps that the proposed system is specifically designed to address. Each gap is grounded in evidence from the studies reviewed in Section 1.2:

- Gap 1 — Single-label datasets cannot support multi-disease detection: As established by Mohanty et al. [3], the PlantVillage dataset assigns a single disease label per image, reflecting controlled inoculation conditions rather than field reality. Kibriya and Rafique [4] and Garba and Jain [5] built effective classifiers on this data but both systems were constrained to single-disease outputs, making them inadequate for real farm scenarios where co-infection is common. No existing mobile system for the Sri Lankan context addresses this co-infection problem through a parallel binary architecture. The proposed three-model parallel binary approach directly resolves this limitation without requiring multi-label ground truth.
- Gap 2 — Grad-CAM explainability has not been delivered to farmers through mobile systems: Drake and Henry [8] confirmed that most XAI implementations in agriculture remain confined to research prototypes and have not been validated with actual farmers. Mahmud et al. [7] demonstrated the technical feasibility of Grad-CAM for tomato disease detection but did not extend their system to a mobile application. No prior Sri Lankan agricultural AI system delivers real-time visual explanations to farmers on a smartphone, leaving a significant trust and transparency deficit that the proposed system addresses.
- Gap 3 — Existing systems provide no verified treatment advisory: Garba and Jain [5] and Jayasinghe et al. [10] both demonstrated that disease detection systems can be effectively built, but neither incorporated a verified advisory component linked to authoritative agricultural sources. Detection without actionable management guidance leaves farmers unable to act on the system's output. The proposed system bridges this gap by integrating DoA HORDI-verified treatment guidelines, with an LLM exclusively employed to reformulate verified content into farmer-accessible language.
- Gap 4 — No prior system uses an LLM as a constrained advisory translator: The use of language models in agricultural AI systems studied in this review was limited to general text generation, which carries risks of factual distortion. No Sri Lanka-focused system employs an LLM in a tightly bounded role restricted to paraphrasing pre-verified DoA content. This represents a novel design contribution that preserves factual authority while substantially improving accessibility for farmers across literacy levels.
- Gap 5 — No integrated system targets the three priority tomato diseases for Sri Lankan farmers: The reviewed systems addressed either general crop disease detection [3], single-

disease tomato classification [4], [5], or Sri Lankan crop diseases in different crop categories [10]. None specifically targets Early Blight, Bacterial Spot, and Tomato Leaf Curl Virus as a prioritised, co-occurring disease set, and none integrates detection, Grad-CAM explainability, and DoA-verified advisory support within a single mobile platform tailored to Sri Lankan agro-climatic conditions. The proposed system is the first to address all five identified gaps in a unified architecture.

System	Target Diseases	Multi-Disease	Grad-CAM XAI	Mobile App	Advisory (DoA-verified)	SL Context
Mohanty et al. [3]	14 crops, 26 diseases	No	No	No	No	No
Mahmud et al. [7]	Tomato (multi-class)	No	Yes	No	No	No
Kibriya & Rafique [4]	Tomato	No	No	No	No	No
Garba & Jain [5]	Tomato	No	No	No	No	No
Jayasinghe et al. [10]	Rice, Vegetables (SL)	No	No	Partial	No	Yes
Proposed System	EB, BS, TLCV (SL)	Yes	Yes	Yes	Yes (DoA+LLM)	Yes

Table 1: Comparison of existing systems against the proposed system

2. OBJECTIVES

2.1 Main Objective

The main objective of this research is to develop an AI-powered mobile application capable of detecting multiple tomato leaf diseases — specifically Early Blight, Bacterial Spot, and Tomato Leaf Curl Virus — simultaneously from a single leaf image using a parallel binary CNN classification architecture, integrated with Grad-CAM-based explainable AI and a DoA-verified advisory system accessible to smallholder farmers in Sri Lanka.

2.2 Specific Objectives

- To design and train three independent binary CNN classifiers, each specialised in detecting a single target disease (Early Blight, Bacterial Spot, or Tomato Leaf Curl Virus), using transfer learning on single-label tomato leaf disease datasets.
- To implement a parallel detection pipeline that simultaneously routes a preprocessed leaf image through all three binary classifiers and combines their outputs to produce a multi-disease detection result with confidence scores.
- To integrate Grad-CAM visual explanation generation for each CNN model so that each prediction is accompanied by a heatmap identifying the leaf regions that most influenced the model's decision.

- To develop an advisory module that retrieves verified treatment and prevention guidelines from the Department of Agriculture Sri Lanka (HORDI) and employs an LLM exclusively to translate these guidelines into clear, farmer-friendly language.
- To build and deploy a cross-platform React Native mobile application through which farmers can capture or upload tomato leaf images and receive detection results, confidence scores, Grad-CAM heatmaps, and advisory recommendations in real time.
- To evaluate the performance of each binary CNN classifier using standard metrics (accuracy, precision, recall, F1-score) and assess the usability and accessibility of the complete system through user acceptance testing with Sri Lankan farmers and agricultural extension officers.

3. METHODOLOGY

This research adopts a Design Science Research (DSR) methodology, which is appropriate for studies that involve the design, construction, and evaluation of innovative IT artefacts intended to solve real-world problems. The research follows an experimental approach for model training and evaluation, combined with iterative Agile development for the mobile application. The methodology is structured across five key phases: deep learning foundations, data preparation, model development, system integration, and evaluation.

3.1 Key Technical Foundations

3.1.1 Convolutional Neural Networks (CNNs)

The proposed detection framework is built upon Convolutional Neural Networks, a category of deep learning architectures whose structural design is inherently suited to the analysis of grid-structured inputs such as digital imagery. These networks interleave convolutional operations that extract localised spatial features using trainable filter banks, downsampling layers that consolidate feature representations while discarding redundant spatial detail, normalisation components, and terminal classification layers. Their utility in plant pathology stems from the spatially localised nature of disease indicators — including lesion geometry, colour deviation, tissue necrosis, and surface texture changes — which are precisely the type of structured visual features that convolutional operations are optimised to detect.

The three binary classifiers proposed in this study will each take a preprocessed leaf image as input and output a binary prediction (1 = disease present, 0 = disease absent) together with a probability score. The architectures will be based on transfer learning foundations to maximise accuracy on the limited per-disease dataset sizes available.

3.1.2 Transfer Learning

By initialising model weights from CNNs pre-conditioned on large repositories such as ImageNet — which encompasses over a million labelled samples across a thousand categories — transfer learning enables domain-specific adaptation through targeted fine-tuning, substantially reducing data and computation requirements [15].

3.1.3 Gradient-weighted Class Activation Mapping (Grad-CAM)

Grad-CAM operates by back-propagating the target class gradient through to the terminal convolutional layer of a trained CNN, using those signals to compute a weighted summation of activation maps that collectively highlight spatially discriminative input regions [11].

3.1.4 Large Language Model for Advisory Translation

The advisory component assigns the LLM a narrowly bounded function: rather than drawing on its internal parameters to generate agronomic recommendations, it receives structured treatment content already validated against DoA HORDI guidelines [2] and reexpresses this material in plain language appropriate for farmers with limited formal education. This design decouples factual authority — retained exclusively by institutional sources — from linguistic accessibility, with the LLM serving solely as a reformulation layer.

3.2 System Architecture

The proposed system architecture consists of three integrated components: the AI detection layer, the explainability layer, and the advisory layer, all orchestrated through a mobile-backend pipeline.

3.2.1 Parallel Binary Detection Pipeline

The defining architectural contribution of this research is the concurrent execution of independent binary classifiers. Upon image submission through the mobile interface, a standardised preprocessing sequence is applied: the image is rescaled to the required input resolution for the target architecture (224×224 pixels for MobileNetV2, EfficientNet, and ResNet), pixel intensities are scaled to the unit interval, and channel statistics are normalised. The prepared image is then dispatched in parallel to all three binary CNN models hosted within the Node.js backend:

- Model 1 — Early Blight Detector: Outputs P (Early Blight present) $\in [0,1]$ and binary prediction.
- Model 2 — Bacterial Spot Detector: Outputs P (Bacterial Spot present) $\in [0,1]$ and binary prediction.
- Model 3 — Tomato Leaf Curl Virus Detector: Outputs P (TLCV present) $\in [0,1]$ and binary prediction.

The combined output of the three models constitutes a multi-disease detection result. For example, if Models 1 and 2 output positive predictions while Model 3 outputs a negative prediction, the system reports Early Blight and Bacterial Spot as detected, and Tomato Leaf Curl Virus as not detected. This architecture enables multi-disease reporting without requiring multi-label training data, overcoming the fundamental limitation of single-label datasets.

3.2.2 Explainability Layer

By visually communicating which leaf areas drove the model's decision, this feature enables users to cross-reference the AI's focus against their own field observations, reinforcing confidence in the system's reliability [7], [8].

3.2.3 Advisory Module

Once a disease is confirmed, the backend queries the advisory repository to retrieve the corresponding DoA-HORDI treatment protocols for the identified condition(s) [2]. The retrieved guidelines — encompassing agronomic management, chemical intervention options, resistant cultivar recommendations, and preventive strategies — are forwarded to the LLM with an explicit instruction set restricting it to reformulation only.

3.2.4 System Deployment Stack

The deployment stack comprises the following components:

- Frontend: React Native cross-platform mobile application (Android and iOS), providing camera capture, image upload, results display (disease list, confidence scores, heatmap images, advisory text), and offline image storage.
- Backend: Node.js server hosting the three CNN models, managing inference requests, triggering Grad-CAM generation, interfacing with the advisory database, and calling the LLM API.
- Model hosting: TensorFlow.js or Python Flask micro-server (accessed via REST API from Node.js) for serving CNN model inference.
- Database: Firebase or SQLite for storing diagnostic history, farmer profiles, and advisory content.
- Cloud: AWS or Google Cloud for scalable model hosting and storage.

3.3 Dataset

The dataset for training and validating each binary classifier will be assembled from the following sources:

- Primary: PlantVillage dataset (open access), which contains single-disease-labeled tomato leaf images including Early Blight and Bacterial Spot. Images will be extracted per disease class to create individual binary datasets (disease-positive vs. healthy).

- Supplementary: Kaggle tomato leaf disease datasets and any other openly licensed repositories providing labeled images of Tomato Leaf Curl Virus. All dataset usage will comply with original licenses and include proper attribution.
- Local field collection (if needed): Images of diseased tomato leaves collected from Sri Lankan cultivation areas (Nuwara Eliya, Badulla, Matale) to improve model generalisation to local cultivar morphology, lighting conditions, and disease phenology.

Data preparation will include:

- Data partitioning: Images will be divided into training (70%), validation (15%), and evaluation (15%) subsets using class-stratified sampling, with source origin tracked to prevent information leakage across splits when integrating multiple datasets.
- Augmentation: Synthetic diversity will be introduced via in-plane rotations, axis-aligned reflections, scale variation, luminance and contrast perturbation, and subtle hue shifts — each bounded to ensure the resulting images remain representative of authentic field capture conditions.
- Quality controls: Images will be screened programmatically for motion blur and exposure artefacts, subject to a minimum short-edge resolution threshold of 512 pixels, and deduplicated via perceptual hashing.
- Annotation: CSV-format labels with fields: `image_id`, `disease_class`, `source_dataset`, `collection_mode` (field/controlled), `augmented` (true/false).

3.4 Software Development Approach

Software development will be governed by an Agile methodology operationalised through the Scrum framework, with work organised into focused two-week iteration cycles. This structure enables progressive delivery, continuous quality verification, and the systematic incorporation of stakeholder input at each development stage. Planned sprint deliverables encompass: dataset curation and classifier training; Grad-CAM implementation and validation; backend API construction; React Native interface development; advisory pipeline integration; and comprehensive end-to-end verification including user acceptance trials.

3.4.1 Requirements Gathering

- Stakeholder interviews and surveys with tomato farmers in disease-prevalent districts (Nuwara Eliya, Badulla, Matale, Polonnaruwa) to identify practical diagnostic needs, language preferences (Sinhala/Tamil), and smartphone usage patterns.
- Consultation with HORDI plant pathologists to validate target disease selection, advisory content accuracy, and Grad-CAM heatmap interpretability.

3.4.2 Feasibility Study

Technical feasibility: All core technologies required for implementation — including TensorFlow/PyTorch for model development, React Native for cross-platform mobile engineering, Node.js for server-side logic, and cloud infrastructure for deployment — are well-established, openly accessible, and extensively documented. The research team has demonstrated proficiency across these toolsets.

Economic feasibility: By drawing on open-source software ecosystems, no-cost development tiers offered by cloud providers, and freely available training datasets, the project substantially constrains its development expenditure. Primary cost obligations are limited to production cloud hosting and platform distribution fees.

Operational feasibility: The intended user base comprises smallholder farmers with basic familiarity with smartphone use. Sinhala and Tamil language support, combined with an interface designed for minimal navigation complexity, will facilitate adoption across the diverse demographic profile of Sri Lanka's farming population.

3.4.3 Implementation Phases

1. Dataset preparation: Curate, preprocess, augment, and partition the training corpus for each of the three binary classifiers.
2. Model training: Fine-tune candidate transfer learning architectures on the prepared datasets and identify the optimal configuration for each disease-specific classifier.
3. Explainability integration: Implement Grad-CAM inference pipelines for each trained model and validate heatmap output quality.
4. Backend construction: Build and deploy the Node.js inference layer with REST endpoints servicing image classification requests, explainability generation, and advisory content retrieval.
5. Frontend development: Engineer the React Native mobile client with integrated image capture, structured results presentation, and persistent diagnostic history.
6. Advisory integration: Connect the DoA-HORDI content store to the LLM translation pipeline and validate end-to-end advisory output quality.
7. Final verification: Execute a comprehensive testing programme covering unit, integration, and system levels, culminating in structured user acceptance trials with target farmer and officer groups.

3.4.4 Testing Strategy

- **Classifier performance:** Each binary model will be assessed against the withheld test partition using a standard set of binary classification metrics, including accuracy, precision, recall, F1-score, and the area under the receiver operating characteristic curve.
- **Multi-disease validation:** Composite outputs from all three classifiers will be evaluated on constructed co-infection scenarios to verify that the parallel pipeline correctly identifies simultaneous disease presence.
- **Explainability assessment:** Generated Grad-CAM visualisations will be reviewed by HORDI plant pathologists to confirm that highlighted regions correspond to pathologically relevant leaf features.
- **Application verification:** The mobile application will be subjected to component-level unit tests, inter-module integration tests, full system tests, and structured field acceptance trials conducted with representative farmers and agricultural extension staff.
- **Data allocation:** A 70/15/15 train-validation-test partition will be maintained independently for each binary classifier.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

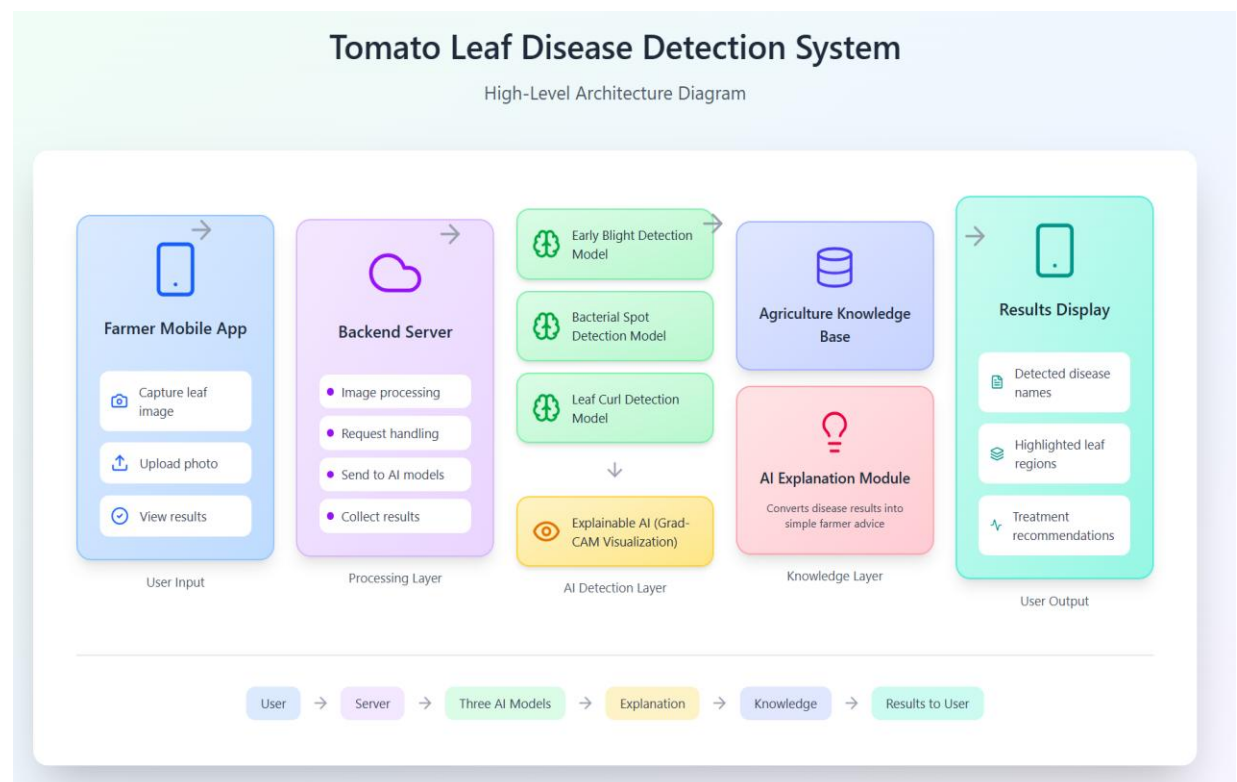


Table 2: System component interaction summary

5. USER REQUIREMENTS

5.1 Stakeholder Identification

- Primary users: Small-scale tomato farmers in Sri Lanka (Nuwara Eliya, Badulla, Matale, Polonnaruwa districts).
- Secondary users: Agricultural extension officers and HORDI field staff.
- Institutional stakeholders: Department of Agriculture Sri Lanka (HORDI), SLIIT, research supervisors.

5.2 Functional Requirements

1. The system shall allow users to capture a tomato leaf image using the device camera or upload from the gallery.
2. The system shall preprocess the image (resize, normalise) and transmit it to the backend server for inference.
3. The system shall independently execute all three binary CNN classifiers on the submitted image.
4. The system shall display which diseases (Early Blight, Bacterial Spot, Tomato Leaf Curl Virus) are detected, along with the confidence score for each.
5. For each detected disease, the system shall generate and display a Grad-CAM heatmap overlaid on the original leaf image.
6. The system shall retrieve and display DoA-verified treatment and prevention advice for each detected disease, in farmer-friendly language.
7. The system shall support bilingual output in Sinhala and Tamil.
8. The system shall allow users to save diagnostic results for future reference.
9. The system shall support offline image storage with synchronisation when connectivity is restored.

5.3 Non-Functional Requirements

Requirement	Specification
Accuracy	Each binary classifier shall achieve $\geq 90\%$ accuracy on the held-out test set
Performance	End-to-end inference and response generation shall complete within 10 seconds
Usability	Interface shall require no technical training; primary navigation within 3 taps
Cross-platform	Application shall function on Android (v8.0+) and iOS (v13.0+)
Reliability	System uptime $\geq 99\%$ during active farming periods (peak seasons)
Security	No personally identifiable information shall be stored without user consent
Accessibility	Font sizes and UI elements shall be accessible for users with low literacy
Localisation	Full Sinhala and Tamil language support for all UI elements and advisory text

Table 3: Non-Functional Requirements

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The primary target audience includes approximately 500 to 1000 small-scale tomato farmers across Sri Lanka's major cultivation districts (Nuwara Eliya, Badulla, Matale, Polonnaruwa). Secondary audiences include district agricultural extension officers, agricultural input retailers, Agri-insurance providers, and research institutions such as HORDI. The Department of Agriculture Sri Lanka represents both a key stakeholder and a potential institutional distribution partner for scaling adoption [2].

6.2 Value Proposition

- Enables instant, on-farm multi-disease detection without requiring an agricultural officer to be physically present.
- Reduces crop losses by enabling timely treatment intervention through early detection.
- Builds farmer trust through Grad-CAM visual explanations of AI predictions.
- Delivers DoA-verified, accessible management advice in Sinhala and Tamil.
- Requires only a smartphone — no additional hardware investment.

6.3 Revenue Model

- Freemium model: Basic disease detection free; premium features (diagnostic history, push notifications, extended advisory) available via subscription (LKR 500/month or LKR 4,500/year).
- Institutional licensing: Annual license for DoA extension services, agrochemical companies, and NGOs for bulk deployment.
- Data insights (privacy-preserving, aggregated): Disease outbreak trend data sold to DoA and crop insurance companies for risk modelling.

6.4 Competitive Advantage

No existing mobile application in the Sri Lankan market offers the combination of: (1) parallel multi-disease detection from single-label training data, (2) Grad-CAM heatmap explainability, and (3) DoA-verified bilingual advisory support. The system's architecture directly addresses the co-infection detection problem unaddressed by all comparable systems identified in the literature review.

6.5 Intellectual Property

The parallel binary classification architecture for multi-disease detection from single-label datasets represents a novel system design that may be protectable as a software patent or registered as intellectual property through SLIIT and the National Intellectual Property Office of Sri Lanka. All

DoA advisory content will be used under appropriate data sharing agreements with the Department of Agriculture.

7. BUDGET AND JUSTIFICATION

Requirement	Cost (LKR)	Justification
Travel expenses for field visits (Nuwara Eliya, Badulla, Matale) data collection and UAT	12,000.00	Transport and accommodation for three district visits for dataset collection and user testing
Cloud hosting (AWS/Google Cloud AI model hosting, storage, compute)	16,000.00/month	Required for hosting three CNN models and Node.js API at sufficient computational capacity
LLM API usage (OpenAI/Gemini advisory translation)	5,000.00/month	API calls for advisory text simplification during development and pilot deployment
Google Play Store (one-time developer registration)	7,542.00 (USD 25)	Required to publish and distribute the Android application
Apple App Store (annual developer account)	29,866.00/year	Required to publish and distribute the iOS application
GPU compute (Google Colab Pro+ for model training)	6,500.00/month	Accelerated training for three CNN models on GPU instances
TOTAL (estimated)	~ 76,908.00	

Table 4: Budget Breakdown

8. WORK BREAKDOWN STRUCTURE (WBS)

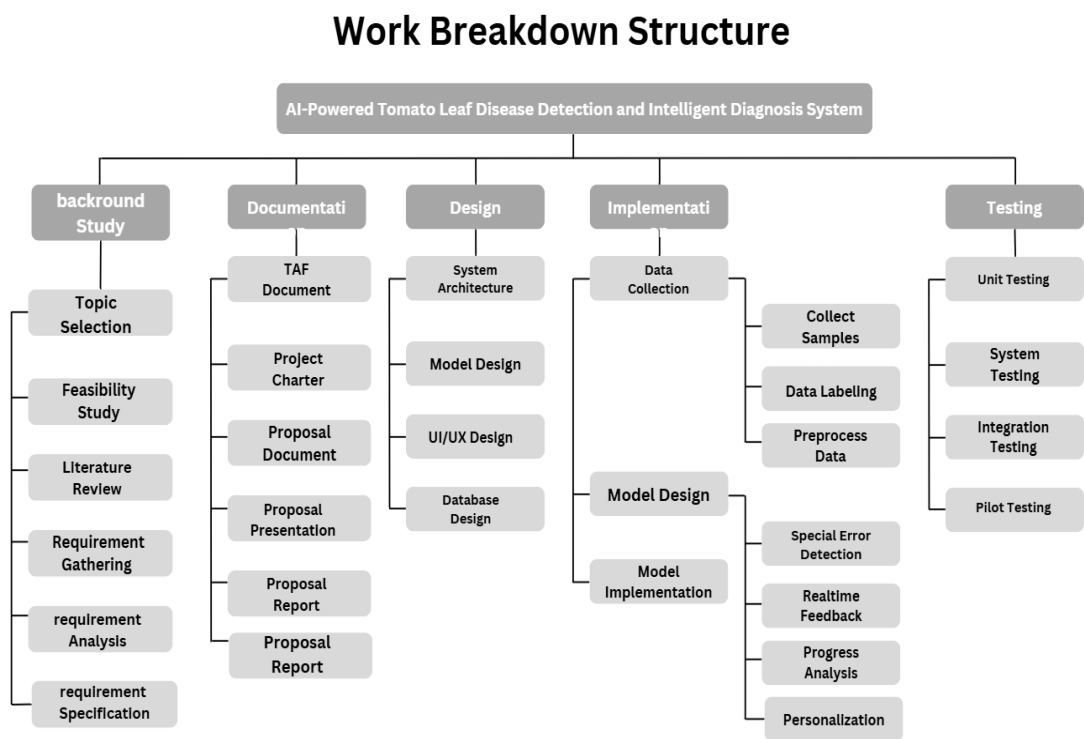


Table 5: Work Breakdown Structure

9. GANTT CHART

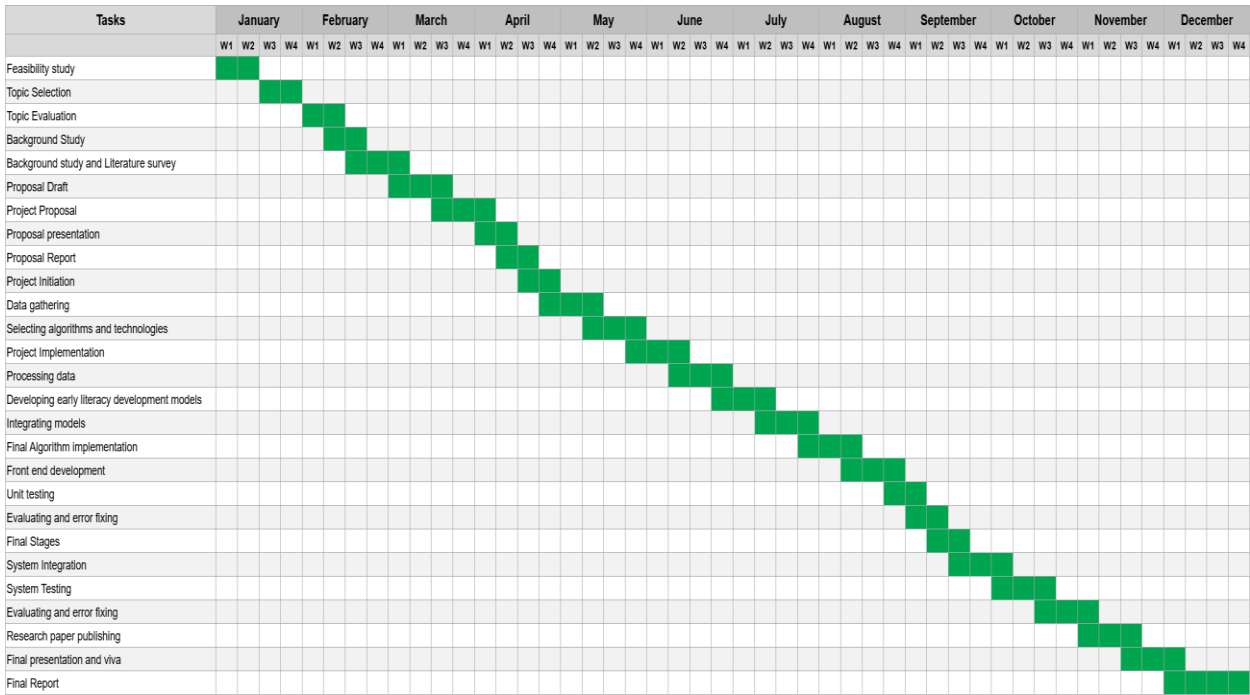


Table 6: Project Gantt Chart (May 2025 – July 2026). Shaded cells indicate active work periods.

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